

Dipankar Saha

Senior Backend Engineer — AI/LLM & Distributed Systems

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Summary

Backend engineer with 6+ years building distributed Node.js/TypeScript systems and AI-powered workflows — event-driven NestJS microservices over REST, GraphQL, and gRPC, RabbitMQ messaging, multi-tenant RBAC, rule and policy enforcement, and production observability. Independently built an 11-service AI notification platform and a multi-agent system with run traces, per-call cost tracking, and human-in-the-loop approvals.

Skills

Backend: Node.js, NestJS, TypeScript, REST APIs, GraphQL, gRPC, JWT / OAuth2, RBAC

Messaging & Architecture: RabbitMQ, Event-Driven Architecture, Microservices Design, Turborepo Monorepo

Databases: PostgreSQL, pgvector, MySQL, MongoDB, Redis, Prisma, Drizzle ORM

Observability: OpenTelemetry, Prometheus, Grafana, Jaeger, Loki

Cloud & DevOps: AWS (EC2, S3, EKS, ECR, ELBv2), Kubernetes, Helm, ArgoCD (GitOps), Terraform, Docker, GitHub Actions, Jenkins, CI/CD

AI & LLM: OpenAI API, RAG (pgvector), LangGraph agents, Multi-Agent Orchestration, Structured Outputs (tool use), LLM Observability (tokens / cost / traces), Human-in-the-Loop Approvals, Prompt Engineering

Frontend: React, Next.js (App Router), Tailwind CSS, Radix UI

Experience

Senior Software Engineer — Veztraa Solutions

Remote | Mar 2023 — Present

- Design and build enterprise SaaS applications (React/Next.js/Node.js/NestJS microservices, REST/GraphQL, event-driven messaging, JWT/RBAC).
- Lead architecture discussions and mentor engineers; own deployment (Docker, Kubernetes on AWS, CI/CD via GitHub Actions/Jenkins) and DB performance (MongoDB, MySQL).

Independent projects (not company-owned):

- **Personal Growth AI OS** — Seven structured LLM agents (Gemini / OpenAI / Anthropic provider ladder with automatic fallback) run on a persisted execution state machine — idempotency keys, cancellation, stale-run reaping, an append-only trace log, and a human approval gate where agent-proposed actions wait for approve/reject — with 70+ runs and 550+ trace events recorded. ([proof](#))
- **AI-Powered Event-Driven Notification Platform** — Multi-tenant RBAC and API-key management; multi-channel delivery with retry scheduling; multi-provider AI event analysis and pgvector RAG duplicate-incident detection. ([proof](#))
- **Platform Infrastructure/Platform GitOps — Terraform provision + Helm + ArgoCD Delivery** — Reusable Terraform modules (network, EKS, ECR, ELBv2 load balancer) provision the AWS runtime for the whole platform. ([proof](#))

Software Engineer — Tata Consultancy Services (TCS)

Kolkata, India | Apr 2021 — Feb 2023

- Built enterprise insurance applications for Zurich Insurance (Angular/Node.js) — scalable REST APIs for high-volume workflows, in an Agile/Scrum team.
- Worked with distributed teams in Agile/Scrum, resolving production issues while keeping availability high.

Tech: Angular, Node.js, JavaScript, REST APIs, MySQL, Git

Software Engineer — Asparrowtech

Indore, India | Jun 2020 — Mar 2021

- Built responsive JavaScript/Node.js web apps and backend APIs, integrating third-party services for client projects.
- Contributed to database design and collaborated directly with clients to deliver custom software.

Tech: JavaScript, Node.js, REST APIs

Projects

- **Japshop Admin Panel — Financial Logbook** — Per-customer GAVE/GOT ledger entries across multiple logbooks, with running balances per customer. ([Case study](#))

Education

B.Tech in Information Technology, Siliguri Institute of Technology — 2016 – 2020